

SOFTWARE ENGINEERING LAB - 1

Lab 1: Requirements Engineering & UML Use-Case Modelling

Team Number: 08

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Problem Statement #08 — Digital Campus Library Reservation Gateway

Target Stakeholders / Actors: Student Member, Head Librarian

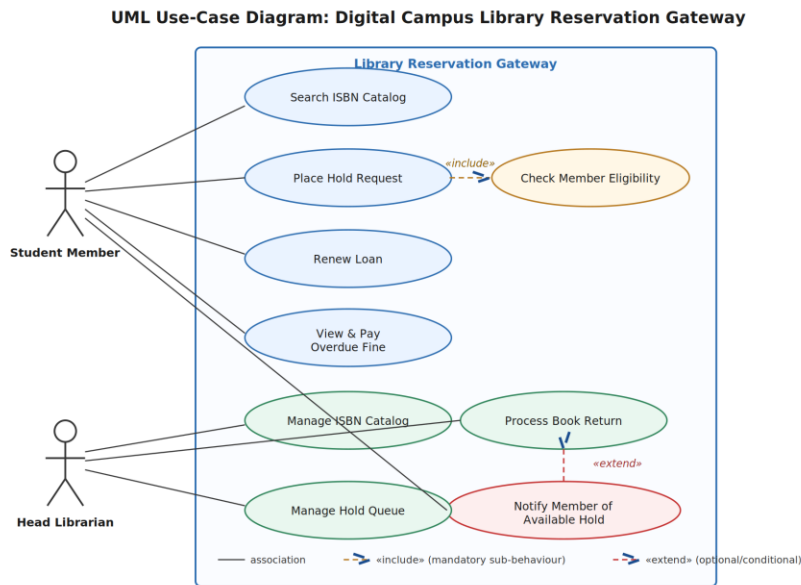
1. Requirements Table

Five Functional Requirements (FR-001–FR-005, with FR-001 as given) and two Non-functional Requirements (NFR-001–NFR-002, with NFR-001 as given).

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow members to search the ISBN repository and place hold requests on currently loaned books to enter a FIFO waitlist.	High	Pass: User is assigned correct position in queue. Fail: User already holding max allowed books places an additional reservation.	Enables equitable, order-based access to high-demand titles without manual librarian intervention.
FR-002	Functional	The system shall automatically notify a member (email/SMS) within 5 minutes of a held book being checked in, and hold the copy for 48 hours before releasing it to the next member in the queue.	High	Pass: Notification sent within 5 minutes and copy auto-released after 48 hours if uncollected. Fail: Notification not sent or copy held indefinitely.	Prevents held copies from sitting idle and ensures timely, fair circulation of returned titles.
FR-003	Functional	The system shall automatically calculate and post an overdue-fine transaction to a member's account for each day a borrowed book remains unreturned past its due date.	High	Pass: Fine amount equals (days overdue x daily rate) and appears in transaction history by end of due date. Fail: No fine posted or incorrect amount calculated.	Automates fine accrual, reducing manual librarian effort and encouraging timely returns.
FR-004	Functional	The system shall allow the Head Librarian to add, update, or deactivate ISBN catalog records, including title metadata and physical copy counts.	Medium	Pass: Catalog change is reflected in member search results within 1 minute of saving. Fail: Change is not persisted or not visible to members.	Keeps the searchable catalog accurate as the physical collection changes over time.
FR-005	Functional	The system shall allow a member to renew a currently borrowed book for one additional loan period, provided no other member holds an active reservation on that title.	Medium	Pass: Due date is extended by the standard loan period and confirmation is shown. Fail: Renewal is blocked when an active hold exists on the title.	Gives members flexibility to keep books longer while protecting the FIFO queue's fairness for waiting members.
NFR-001	Non-functional	The catalog search query shall return paginated results for 50,000 titles in less than 150 ms.	High	Pass: Benchmarking tests confirm target latency and security standards are met under simulated peak load.	Ensures a responsive search experience during peak enrollment/exam periods when catalog usage spikes.
NFR-002	Non-functional	The system shall be available 99.9% of the time during the academic semester (excluding scheduled maintenance) and shall automatically recover from a single-node failure within 5 minutes.	High	Pass: Uptime monitoring confirms $\geq 99.9\%$ availability over a rolling 30-day window and failover tests recover service within 5 minutes. Fail: Downtime exceeds threshold or requires manual recovery.	Hold placement, renewal, and fine payment are time-sensitive actions members rely on throughout the day.

2. UML Use-Case Diagram

Actors: Student Member and Head Librarian. The diagram below shows all primary use cases along with the modelled «include» and «extend» relationships.



Include relationship: "Place Hold Request" includes "Check Member Eligibility" (eligibility is always verified before a hold is created).

Extend relationship: "Notify Member of Available Hold" extends "Process Book Return" (only triggered when the returned title has an active hold queue).

3. Use-Case Flow Specification

Use Case Name	Notify Member of Available Hold
Related Requirement	FR-002 (extends Process Book Return)
Primary Actor	Student Member (recipient) Secondary Actor: Head Librarian (via Process Book Return)
Preconditions	1) A returned book has one or more members waiting in its FIFO hold queue. 2) The Head Librarian has just checked in the returned copy.
Postconditions	Success: The next member in the queue is notified and the copy is reserved for them for 48 hours. Failure: The notification is not delivered, or the copy is released without notifying the intended member.

Main Success Scenario

- Head Librarian checks in the returned book (base use case: Process Book Return).
- System detects an active hold queue on the title at the extension point.
- System identifies the member at position 1 in the FIFO queue.
- System reserves the physical copy for that member and starts a 48-hour hold timer.
- System sends an email/SMS notification informing the member the book is ready for pickup.
- System updates the member's dashboard to show "Ready for Pickup".
- Use case ends, completing the extension of Process Book Return.

Alternate Flow A1: Member Does Not Collect Within 48 Hours

- A1.1. At step 4, the 48-hour hold timer expires before the member checks out the book.
- A1.2. System releases the reserved copy back into circulation.
- A1.3. System removes the expired member's hold record and notifies them that the hold has lapsed.
- A1.4. System repeats steps 3-6 for the next member in the queue, if one exists.
- A1.5. Use case ends.